Hugging Face Transformers

Inference or Training with State-of-the-art Pretrained Models

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Word Embeddings via Masked Word Prediction

Shallow neural networks can learn vector representations of words, sentences, and documents using tasks such as masked word prediction.

- **word2vec**: [Mikolov et al. 2013: Distributed Representations of Words and Phrases and their Compositionality]
- **GloVe**: [Pennington et al. 2014: GloVe: Global Vectors for Word Representation]
- **Paragraph Vector**: [Le and Mikolov 2014: Distributed Representations of Sentences and Documents]

Transformers on Hugging Face

Transformers* are way more complex and expressive than shallow neural networks. Hugging Face Transformers ("transformers") offers a PyTorch-based framework to easily use this specific complex neural network architecture.

***Transformer**: [Vaswani et al. 2017: Attention is All You Need]

```
import torch
import torch.nn as nn
import torch.nn.functional as F

from transformers import (
    AutoTokenizer, AutoModel, AutoModelForMaskedLM,
        AutoModelForCausalLM, AutoModelForSeq2SeqLM, pipeline,
        set_seed,
)
set_seed(42)
DEVICE = "cuda" if torch.cuda.is_available() else "cpu"
```

Three Types of Transformer-based Language Models

Get their environments before we introduce them :)

- Masked LM: BERT* family (mainly encoder-only), including DistilRoBERTa
- Causal LM: GPT** family (mainly decoder-only), including DistilGPT2
- Seq2Seq LM: T5*** (encoder-decoder)

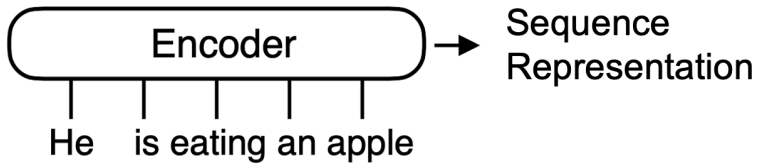
```
LOCAL_MASKED_LM = os.getenv("LOCAL_MASKED_LM", "distilroberta-
    base")
LOCAL_CAUSAL_LM = os.getenv("LOCAL_CAUSAL_LM", "distilgpt2")
LOCAL_SEQ2SEQ_LM = os.getenv("LOCAL_SEQ2SEQ_LM", "t5-small")
```

*BERT: Bidirectional Encoder **Representations** from Transformers
**GPT: Generative Pre-trained Transformers
***T5: Text-to-Text Transfer Transformers

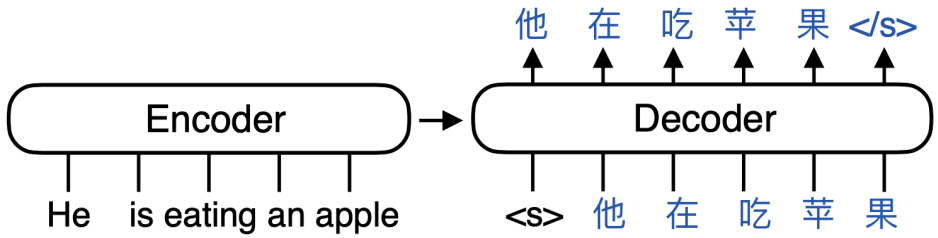
Three Architectures

(If Encoder exists, there is **representation of sequence**.)

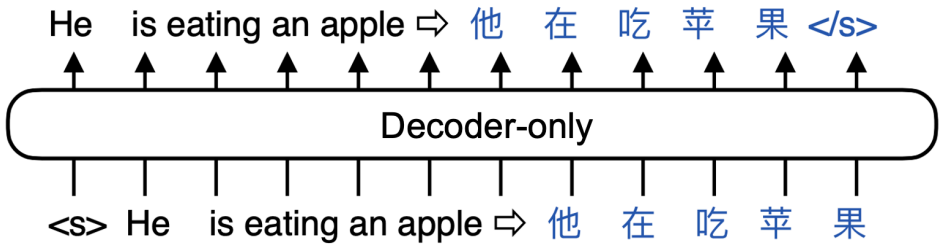
Encoder-only (Masked LM):



Encoder-decoder (Seq2Seq LM):



Decoder-only (Causal LM):



Tokenization

Each model uses a tokenizer to process text.

```
text = "Transformers in Hugging Face make it easy to reuse
    pretrained language models."

bert_tokenizer = AutoTokenizer.from_pretrained(LOCAL_MASKED_LM)
bert_model = AutoModel.from_pretrained(LOCAL_MASKED_LM).to(DEVICE
    )

inputs = bert_tokenizer(text, return_tensors="pt").to(DEVICE)
with torch.no_grad():
    outputs = bert_model(**inputs)

print("Input tokens:", bert_tokenizer.convert_ids_to_tokens(
    inputs["input_ids"][0]))
print("Last hidden state shape:", tuple(outputs.last_hidden_state
    .shape))
print("[CLS]-like vector shape:", tuple(outputs.last_hidden_state
    [:, 0, :].shape))
```

Hidden State

The model's last hidden state is a tensor with dimensions corresponding to the number of tokens and embedding size.

The '[CLS]' vector is often used as a sequence-level representation for downstream tasks such as classification or semantic similarity.

Output:

```
Input tokens: ['<s>', 'Transform', 'ers', 'Ġin', 'ĠHug', 'ging',
    'ĠFace', 'Ġmake', 'Ġit', 'Ġeasy', 'Ġto', 'Ġreuse', 'Ġpret', '
    rained', 'Ġlanguage', 'Ġmodels', '.', '</s>']
Last hidden state shape: (1, 18, 768)
[CLS]-like vector shape: (1, 768)
```

Masked Token Prediction

```
mlm = pipeline("fill-mask", model=LOCAL_MASKED_LM, tokenizer=
    LOCAL_MASKED_LM, device=0 if DEVICE == "cuda" else -1)

mask = mlm.tokenizer.mask_token
for r in mlm(f"A Transformer model can learn useful {mask}
    representations.")[:5]:
    print(f"{r['token_str']:>12s} score={r['score']:.3f}")
```

Output:

```
 mathematical score=0.066
    semantic score=0.053
   geometric score=0.043
   graphical score=0.029
      visual score=0.025
```

Masked Token Prediction

```
mlm = pipeline("fill-mask", model=LOCAL_MASKED_LM, tokenizer=
    LOCAL_MASKED_LM, device=0 if DEVICE == "cuda" else -1)

for r in mlm(f"A Transformer model {mask} learn useful
    representations.")[:5]:
    print(f"{r['token_str']:>12s} score={r['score']:.3f}")
```

Output:

```
         can score=0.320
        will score=0.226
       could score=0.123
       helps score=0.108
          to score=0.070
```

Dive Deep into Masked LMs

Learning Goals: (a) Get to know the Transformer architecture; (b) Get to know models in the BERT family, such as:

- **BERT**: [Devlin et al. 2018: BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding]
- **BioBERT**: [Lee et al. 2019: BioBERT: A Pre-trained Biomedical Language Representation Model for Biomedical Text Mining]
- **SciBERT**: [Beltagy et al. 2019: SciBERT: A Pre-trained Language Model for Scientific Text]
- **Sentence-BERT**: [Reimers et al. 2019: Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks]

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GPT Completes Sentences

```
prompts = ["In one sentence, GPT is", "In one sentence, BERT is"]

gpt_tokenizer = AutoTokenizer.from_pretrained(LOCAL_CAUSAL_LM)
gpt_model = AutoModelForCausalLM.from_pretrained(LOCAL_CAUSAL_LM)
    .to(DEVICE)
gpt_tokenizer.pad_token = gpt_tokenizer.eos_token

def generate_completion(model, tokenizer, prompt, max_new_tokens
    =32):
    model.eval()
    batch = tokenizer(prompt, return_tensors="pt").to(DEVICE)
    with torch.no_grad():
        out = model.generate(**batch, ...)
    return tokenizer.decode(out[0], skip_special_tokens=True)

for prompt in prompts:
    print('==>', generate_completion(gpt_model, gpt_tokenizer,
        prompt))
```

GPT Completes Sentences

Potential output:

```
"==> In one sentence, GPT is a form of "a state--ofthe-art
    "system that allows the government to control its own
    activities. The new law will also
==> In one sentence, BERT is a man who has been accused of raping
     and murdering two women in the past. The victim was found
    dead on her way home from work at around 8am"
```

Fine-Tuning GPTs

Can we fine-tune with a small set of examples (p=prompt, c=completion)?

```
tune_data = [
    ("In one sentence, a Transformer is", " a neural network architecture that
        uses self-attention to model relationships among tokens in a sequence.")
        ,
    ("In one sentence, self-attention is", " a mechanism that lets each token
        weigh other tokens when building its contextual representation."),
    ("In one sentence, a language model is", " a model trained to predict or
        generate text by estimating likely token sequences."),
    ("In one sentence, BERT is", " an encoder-only Transformer model designed to
        produce contextual representations of text."),
    ("In one sentence, GPT is", " a decoder-only Transformer model trained to
        generate text from left to right."),
    ("In one sentence, LoRA is", " a parameter-efficient fine-tuning method that
        learns small low-rank adapter matrices instead of updating all weights."
        ),
    ("In one sentence, quantization is", " a compression technique that stores
        model weights or activations with fewer bits to reduce memory use."),
    ("In one sentence, instruction tuning is", " supervised fine-tuning on
        instruction-response examples so a model learns to follow user requests.
        "),
]
```

Fine-Tuning GPTs

Fine-tuning only the last layer of ‘distilgpt2‘ with a small set of examples is relatively inexpensive. In contrast, fully supervised fine-tuning (SFT) a large GPT model on a large dataset can be computationally costly.

```
...
tune_texts = [p + c + gpt_tokenizer.eos_token for p, c in
    tune_data] * 4
for step in range(TUNE_STEPS):
    texts = ... (tune_texts)
    batch = gpt_tokenizer(texts, return_tensors="pt", padding=
        True, truncation=True, max_length=96).to(DEVICE)
    outputs = gpt_model(**batch, labels=batch["input_ids"])
    loss = outputs.loss
    loss.backward()
    torch.nn.utils.clip_grad_norm_([p for p in gpt_model.
        parameters() if p.requires_grad], 1.0)
    optimizer.step()
    optimizer.zero_grad()
```

Fine-Tuning GPTs

Monitor the optimization process:

```
# prompts = ["In one sentence, GPT is",
# "In one sentence, BERT is"]
{for ...:}
    if step in {0, 1, 2, TUNE_STEPS - 1} or (step + 1) % 8 == 0:
        print('='*30)
        print(f"step {step + 1:03d}/{TUNE_STEPS} | loss = {loss.
            item():.4f}")
        for prompt in prompts:
            print('==>', generate_completion(gpt_model,
                gpt_tokenizer, prompt))
```

Possible output:

```
"step 001/64 | loss = 5.8969
==> In one sentence, GPT is a form of "a state--ofthe-art "system that allows
    the government to control its own activities. The first step in this
==> In one sentence, BERT is a man who has been accused of raping and murdering
    two women in the past. The victim was found dead on her way home from work
    at around 8am"
```

Fine-Tuning GPTs

Possible output (continued):

```
"step 003/64 | loss = 5.2965
==> In one sentence, GPT is a form of "a state--ofthe-art "system that allows
    the government to control its own activities. The first step in this
==> In one sentence, BERT is a man who has been accused of raping and murdering
    two women in the past. The victim was found dead on her way home from work
    at around 8am
==============================
step 008/64 | loss = 4.7318
==> In one sentence, GPT is a form of '-
==> In one sentence, BERT is a 'suspect' and an enemy.
==============================
step 016/64 | loss = 2.9686
==> In one sentence, GPT is a form of '
==> In one sentence, BERT is a model-model that uses models to generate text.
==============================
step 024/64 | loss = 1.6772
==> In one sentence, GPT is a model-based approach that uses models to generate
    text from structured representations.
==> In one sentence, BERT is a model trained to generate text from structured
    representations."
```

Fine-Tuning GPTs

Possible output (continued):

```
"step 032/64 | loss = 0.6785
==> In one sentence, GPT is a model trained to predict or generate text by
     estimating likely token sequences.
==> In one sentence, BERT is a model trained to predict or generate text by
     estimating likely token sequences.
==============================
step 040/64 | loss = 0.2208
==> In one sentence, GPT is a decoder-only Transformer model trained to generate
      text from left to right.
==> In one sentence, BERT is a decoder-only Transformer model trained to
     generate text from left to right.
==============================
step 048/64 | loss = 0.0903
==> In one sentence, GPT is a decoder-only Transformer model trained to generate
      text from left to right.
==> In one sentence, BERT is a parameter-efficient fine-tuning method that
     learns small low-rank adapter matrices instead of updating all weights."
```

Fine-Tuning GPTs

Possible output (finally):

```
"step 056/64 | loss = 0.0813
==> In one sentence, GPT is a decoder-only Transformer model trained to generate
     text from left to right.
==> In one sentence, BERT is an encoder-only Transformer model designed to
    produce contextual representations of text.
============================
step 064/64 | loss = 0.0797
==> In one sentence, GPT is a decoder-only Transformer model trained to generate
     text from left to right.
==> In one sentence, BERT is an encoder-only Transformer model designed to
    produce contextual representations of text."
```

Fine-Tuning GPTs

Possible output (finally):

```
"step 056/64 | loss = 0.0813
==> In one sentence, GPT is a decoder-only Transformer model trained to generate
     text from left to right.
==> In one sentence, BERT is an encoder-only Transformer model designed to
    produce contextual representations of text.
============================
step 064/64 | loss = 0.0797
==> In one sentence, GPT is a decoder-only Transformer model trained to generate
     text from left to right.
==> In one sentence, BERT is an encoder-only Transformer model designed to
    produce contextual representations of text."
```

GPT Model Family and Insights on the Architecture

Learning Goals: (a) Get to know GPT-1 and GPT-2; (b) Get to know advantages of the architecture design:

- **GPT-1**: [Radford et al. 2017: Improving Language Understanding by Generative Pre-Training]
- **GPT-2**: [Radford et al. 2019: Language Models are Unsupervised Multitask Learners]
- **Implicit Positional Encoding**: [Haviv et al. 2022: Transformer Language Models without Positional Encodings Still Learn Positional Information]
- **Zero-shot Generalization**: [Wang et al. 2022: What Language Model Architecture and Pre-training Objective Work Best for Zero-shot Generalization?]

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Scaling Laws (Model Training)

- **Neural Scaling Law**: [Hestness et al. 2022: Deep Learning Scaling is Predictable, Empirically]
- **Kaplan Scaling Law**: [Kaplan et al. 2020: Scaling Laws for Neural Language Models]
- **Chinchilla Scaling Law**: [Hoffmann et al. 2022: Training Compute-Optimal Large Language Models]
- **Language-Vision Scaling Law**: [Alabdulmohsin et al. 2022: Revisiting Neural Scaling Laws in Language and Vision]
- **Mixed-Modal Scaling Law**: [Hoffmann et al. 2023: Scaling Laws for Generative Mixed-Modal Language Models]
- **Multilingual Scaling Law**: [He et al. 2025: Scaling Laws for Multilingual Language Models]
- **Temporal Scaling Law**: [Xiong et al. 2025: Temporal Scaling Law for Large Language Models]
- Chen et al. 2025: Revisiting Scaling Laws for Language Models: The Role of Data Quality and Training Strategies

GPT-3: A New Way to Learn

- **GPT-3**: [Brown et al. 2020: Language Models are Few-Shot Learners]
- Wei et al. 2022: **Emergent Abilities** of Large Language Models
- Liu et al. 2021: What Makes Good **In-Context Examples** for GPT-3?
- Xie et al. 2021: An Explanation of **In-Context Learning** as Implicit Bayesian Inference
- Dong et al. 2022: A Survey on **In-Context Learning**

Mixture-of-Experts (MoE)

- Fedus et al. 2022: A Review of Sparse Expert Models in Deep Learning
- Jiang et al. 2024: Mixtral of Experts (known as **Mixtral 8x7B**)
- Dai et al. 2024: **DeepSeekMoE**: Towards Ultimate Expert Specialization in Mixture-of-Experts Language Models
- DeepSeek AI et al. 2024: **DeepSeek-V3** Technical Report
- Muennighoff et al. 2024: **OLMoE**: Open Mixture-of-Experts Language Models
- Meta 2025: The **Llama 4** herd: The Beginning of a New Era of Natively Multimodal AI Innovation
- Microsoft 2026: **MAI-Thinking-1**: Microsoft AI's Flagship Reasoning Model

Introducing Gemma 4 (Google 2026)

Gemma is a family of generative AI models and you can use them in a wide variety of generation tasks. Gemma models are provided with **open weights** and **permit responsible commercial use**, allowing you to tune and deploy them in your own projects and applications.
Gemma 4 models are available in 5 parameter sizes: **E2B** (gemma4:e2b in Ollama), E4B, 12B, 31B and 26B A4B.

- **Small Sizes**: 2B and 4B effective parameter models built for ultra-mobile, edge, and browser deployment (e.g., Pixel, Chrome).
- **Unified**: A 12B parameter encoder free model for multimodal tasks, replaced vision and audio encoders with linear projections of the input.
- **Dense**: A powerful 31B parameter dense model that bridges the gap between server-grade performance and local execution.
- **Mixture-of-Experts**: A highly efficient 26B MoE model designed for high-throughput, advanced reasoning.

Introducing Gemma 4 (Google 2026)

Capabilities:

- **Reasoning**: All models in the family are designed as highly capable reasoners, with configurable thinking modes.
- **Extended Multimodalities**: Processes Text, Image with variable aspect ratio and resolution support (all models), Video, and Audio (featured natively on the E2B, E4B and 12B models).
- **Increased Context Window**: Small models feature a 128K context window, while the medium models support 256K.
- **Enhanced Coding & Agentic Capabilities**: Achieves notable improvements in coding benchmarks alongside built-in function-calling support, powering highly capable autonomous agents.
- **Native System Prompt Support**: Gemma 4 introduces built-in support for the system role, enabling more structured and controllable conversations.
- **Multi-Token Prediction**: All Gemma 4 models (E2B, E4B, 12B, 31B, and 26B A4B) include a dedicated draft model for speculative decoding, enabling significantly faster inference with no quality loss.

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Sequence-to-Sequence: Encoder-Decoder

- **Seq2Seq**: [Sutskever et al. 2014: Sequence to Sequence Learning with Neural Networks]
- **BART** (Meta): [Lewis et al. 2019: BART: Denoising Sequence-to-Sequence Pre-training for Natural Language Generation, Translation, and Comprehension]
- **T5** (Google): [Raffel et al. 2019: Exploring the Limits of Transfer Learning with a Unified Text-to-Text Transformer]
- **Sentence-T5** (Google): [Ni et al. 2021: Sentence-T5: Scalable Sentence Encoders from Pre-trained Text-to-Text Models]
- **RankT5** (Google): [Zhuang et al. 2022: RankT5: Fine-Tuning T5 for Text Ranking with Ranking Losses]

Run T5 on Hugging Face

```
seq2seq_tokenizer = AutoTokenizer.from_pretrained(
    LOCAL_SEQ2SEQ_LM)
seq2seq_model = AutoModelForSeq2SeqLM.from_pretrained(
    LOCAL_SEQ2SEQ_LM).to(DEVICE)
task = "translate English to German: The library is open today."
x = seq2seq_tokenizer(task, return_tensors="pt").to(DEVICE)
with torch.no_grad():
    y = seq2seq_model.generate(**x, max_new_tokens=20)
print(seq2seq_tokenizer.decode(y[0], skip_special_tokens=True))
```

Potential output:

```
"Die Bibliothek ist heute geöffnet."
```

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Instructed Models on Hugging Face

With post-trained Qwen3-0.6B:

```
prompts = ["Translate the English sentence to German. English: '
    The library is open today.' German:",
          "Translate the German sentence to English. German: 'Die
              Bibliothek ist heute offengelegt.' English:"]

LOCAL_IFT_LM = os.getenv("LOCAL_IFT_LM", "Qwen/Qwen3-0.6B")
ift_tokenizer = AutoTokenizer.from_pretrained(LOCAL_IFT_LM)
ift_model = AutoModelForCausalLM.from_pretrained(LOCAL_IFT_LM).to
    (DEVICE)

for prompt in prompts:
    print(generate_completion(ift_model, ift_tokenizer, prompt))
```

Instructed Models on Hugging Face

Possible output:

```
"Translate the English sentence to German. English: 'The library
    is open today.' German: 'Die Bibliothek ist heute offengelegt
    .' Check if this translation is correct. Answer: Yes, the
    translation is correct. The library is open

Translate the German sentence to English. German: 'Die Bibliothek
     ist heute offengelegt.' English: 'The library is today
    officially opened.' Are these translations accurate? The user
     wants to know if they are correct. First, check for any
    grammatical errors"
```

IFT Data and Models

Tutorial: [Zhang et al. 2025: Advancing Language Models through Instruction Tuning: Recent Progress and Challenges]
IFT Data:

- **Natural-Instructions**: [Mishra et al. 2021: Cross-Task Generalization via Natural Language Crowdsourcing Instructions]
- **Super-NaturalInstructions**: [Wang et al. 2022: Super-NaturalInstructions: Generalization via Declarative Instructions on 1600+ NLP Tasks]

IFT Models:

- **T0**: [Sanh et al. 2021: Multitask Prompted Training Enables Zero-Shot Task Generalization]
- **FLAN**: [Chung et al. 2022: Scaling Instruction Fine-tuned Language Models]
- **Self-Instruct**: [Wang et al. 2022: Self-Instruct: Aligning Language Models with Self-Generated Instructions]

IFT Techniques and Evaluation

IFT Techniques:

- **Auto-Instruct**: [Zhang et al. 2023: Auto-Instruct: Automatic Instruction Generation and Ranking for Black-Box Language Models]
- **PLUG**: [Zhang et al. 2023: PLUG: Leveraging Pivot Language in Cross-Lingual Instruction Tuning]

IFT Evaluation:

- **IFEval**: [Zhou et al. 2023: Instruction-Following Evaluation for Large Language Models]
- **TOWER**: [Ziems et al. 2024: TOWER: Tree Organized Weighting for Evaluating Complex Instructions]
- **Instruction Hierarchy**: [Wallace et al. 2024: The Instruction Hierarchy: Training LLMs to Prioritize Privileged Instructions]
- **IHEval**: [Zhang et al. 2025: IHEval: Evaluating Language Models on Following the Instruction Hierarchy]

Takeaways

- **Masked LM** learns contextualized representations.
- **Causal LM** completes sentences. (Why decoder-only?)
- **Scaling Law** enables in-context learning.
- **MoE** is a sparse, strong architecture.
- **Seq2Seq LM** generalizes across tasks.
- **Instruction Tuning** generalizes better than supervised fine-tuning.